Across the Channel: Social Networks and the Cross-Border Economic Effects of Brexit

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February 26, 2026

Abstract

Do economic shocks propagate across national borders through social networks? I exploit the Brexit referendum as a quasi-natural experiment and measure cross-border social ties using Facebook's Social Connectedness Index between 96 French départements and 183 UK regions. French départements with stronger social connections to the UK experienced relative housing price *appreciation* after 2016 ($\beta = 0.025$, $p = 0.031$), driven entirely by houses ($\beta = 0.035$) with null effects on apartments—consistent with UK residential buyers, not generic capital flows. Effects concentrate in Channel-facing départements and reverse sign in traditional UK expat areas, suggesting demand reallocation within France. A German placebo partly reflects a broader European integration channel, but permutation inference ($p = 0.038$), property type heterogeneity, and the Swiss franc positive placebo support a UK-specific network transmission mechanism.

JEL Codes: F15, R31, L14, D85

Keywords: Social networks, Brexit, cross-border spillovers, housing markets, Social Connectedness Index

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1. Introduction

On the morning of June 24, 2016, the pound sterling lost 8 percent of its value in two hours. The British electorate had voted to leave the European Union, and financial markets—which had priced in a comfortable Remain victory—scrambled to reprice a continent's worth of economic relationships. The immediate question was what Brexit would cost the United Kingdom. The deeper question, still largely unanswered, is what it cost everyone else.

This paper asks whether economic shocks transmit across national borders through social networks. I study this question using Brexit as a quasi-natural experiment and Facebook's Social Connectedness Index (SCI) to measure the density of cross-border social ties between French départements and British regions. The central finding is that French départements with stronger pre-existing social connections to the UK experienced differential *appreciation* in housing prices after the referendum. A one standard deviation increase in log SCI-weighted UK exposure is associated with a 0.7 percent increase in housing prices per square meter, an effect that emerges at the referendum date and persists through the sample period, with no additional amplification at the 2021 transition.

The sign of the effect is surprising and informative. Rather than transmitting economic distress—the prior expectation—social networks appear to have channeled *demand reallocation* from the UK to France. The Brexit vote triggered a sharp depreciation of sterling, making continental European property relatively cheaper for UK buyers holding euro-denominated assets or income. Simultaneously, the prospective loss of EU free movement rights created urgency among UK nationals to purchase property in France—securing a foothold for continued European residence. Social networks determined *where* this demand landed: départements with denser friendship ties to the UK attracted more of the redirected demand.

The identification strategy exploits within-France variation in social connectivity to the United Kingdom. Two French départements facing identical domestic macroeconomic conditions, identical national fiscal and monetary policy, and identical EU membership status may nevertheless differ sharply in their exposure to the British economy through the density of their cross-border social ties. Creuse, Charente, and Haute-Savoie—départements with strong historical ties to UK expatriate communities—are far more connected to the UK than interior départements with minimal cross-Channel links. When Brexit redirects UK demand toward familiar continental locations, these socially connected départements experience differential price appreciation.

The empirical specification is a continuous-treatment difference-in-differences design. The "treatment" is pre-Brexit SCI intensity—the density of Facebook friendship links from each French département to all 183 UK GADM2 regions, measured from the October 2021 vintage

and reflecting social structures that predate the referendum by years or decades. Départements with higher SCI to the UK are more "treated" by Brexit; the post-referendum indicator captures the timing. The estimating equation interacts log SCI with a post-referendum dummy, controlling for département and quarter-year fixed effects. This design draws on the logic of shift-share instruments (**??**)—the SCI aggregates bilateral friendship links across many UK regions, providing exposure variation analogous to Bartik shares—though the specification is reduced-form rather than instrumental variables. The identifying variation comes from the cross-sectional distribution of social ties: conditional on département and time fixed effects, the only channel through which SCI-weighted UK exposure affects French local outcomes should be through the social network connection itself.

The result is robust to a battery of tests, though a significant German placebo warrants careful interpretation. Permutation inference ($p = 0.038$) confirms the UK estimate lies in the tail of the randomization distribution. A Swiss franc placebo validates the SCI: when the Swiss National Bank abandoned the EUR/CHF floor in January 2015, socially connected départements experienced differential price effects ($\beta = 0.015$, $p < 0.05$). However, an identically constructed German exposure measure is also significant ($\beta = 0.045$, $p = 0.008$)—a result I address with three identification-strengthening tests. Controlling for general European openness preserves the UK effect; residualizing UK SCI on German SCI attenuates it. The mechanism evidence helps adjudicate: effects concentrate entirely in houses (not apartments) and in Channel-facing départements, patterns difficult to explain through generic international connectivity.

A key methodological contribution is the distinction between population-weighted and probability-weighted SCI, following the innovation introduced in **?** and applied to US labor markets in recent work. Population-weighted SCI captures the absolute volume of cross-border social connections (how many friends a French département has in the UK), while probability-weighted SCI captures the share (what fraction of a département's social network is British). In practice, both weighting schemes produce nearly identical coefficients ($\beta = 0.025$ for both), suggesting that the cross-sectional variation in total and mean SCI is highly correlated at the département level.

This paper contributes to three literatures. First, it extends the economics of social networks (**???**) from cross-sectional correlations to causal identification of cross-border transmission. **?** showed that socially connected countries trade more; I show that social connections also channel demand reallocation across borders when political shocks disrupt economic arrangements. **?**, in a foundational contribution, demonstrated that West German regions with more social ties to East Germany experienced higher income growth after reunification. My paper applies the same network logic to a cross-border setting where

the shock (Brexit) redirected demand from the UK toward socially connected continental locations.

Second, the paper contributes to the growing literature on the economic consequences of Brexit (**????**). Existing work has documented aggregate UK GDP losses (**?**), firm-level investment responses (**?**), and within-UK regional variation (**?**). **?** used the SCI to study how trade shock exposure in socially connected UK regions affected Brexit voting patterns—but within the UK and for political rather than economic outcomes. My paper is the first to document how Brexit generated cross-border spillovers to a neighboring country's housing markets—and the finding that these spillovers are *positive* (demand reallocation rather than contagion) adds a new dimension to the debate over Brexit's economic geography.

Third, the paper speaks to the literature on the geography of economic shocks and the channels through which they propagate (**???**). The "China Shock" literature (**?**) showed that trade exposure affects local labor markets; **?** showed that housing wealth shocks propagate through the real economy. I add social networks as a mechanism for cross-border *demand reallocation*—one that operates across national borders and can generate positive spillovers in receiving regions even when the originating shock is negative.

2. Institutional Background

2.1 Brexit: Timeline and Economic Channels

The United Kingdom's withdrawal from the European Union unfolded over a five-year period, with multiple distinct shocks affecting economic conditions at different times and through different channels.

The **referendum** on June 23, 2016 was the first and most dramatic shock. Betting markets had placed the probability of Leave at roughly 30 percent the evening before the vote; financial markets were pricing in a comfortable Remain victory. When the Leave result became clear in the early hours of June 24, the pound sterling fell from $1.50 to $1.33 against the dollar—a 12 percent depreciation that would persist and deepen over subsequent months (**?**). The FTSE 250, an index of domestically focused UK firms, fell 7 percent in a single day. The referendum generated a massive, sudden, and largely unexpected uncertainty shock about the future of UK-EU economic relations.

Article 50 was triggered on March 29, 2017, formally beginning the two-year countdown to withdrawal. This committed the UK to leaving but left the terms of departure entirely uncertain—no trade agreement, no customs arrangement, no framework for financial services or labor mobility had been negotiated. The period from March 2017 through October 2019 was characterized by escalating political uncertainty, with multiple failed parliamentary votes

on the Withdrawal Agreement, two extensions of the Article 50 deadline, and widespread contingency planning for a "no-deal" exit.

The **Withdrawal Agreement** was ratified in January 2020, and the UK formally left the EU on January 31, 2020. However, a transition period preserved the status quo—the UK remained in the single market and customs union—through December 31, 2020. The real **trade friction shock** arrived on January 1, 2021, when new customs requirements, border checks, regulatory divergence, and documentation burdens took immediate effect. Even under the UK-EU Trade and Cooperation Agreement, which preserved zero tariffs on goods, non-tariff barriers imposed substantial new costs on cross-Channel commerce (**?**).

2.2 France–UK Economic Ties

France and the United Kingdom are bound by dense economic, social, and geographic ties. The Channel Tunnel, opened in 1994, physically connects the two countries; the Dover–Calais corridor handles approximately 17 percent of UK trade by value. In 2015, France was the UK's fifth-largest export market and seventh-largest import source.

These aggregate figures mask enormous geographic heterogeneity within France. Île-de-France, the Paris region, hosts the largest British expatriate community in continental Europe—an estimated 200,000–400,000 UK nationals resided in France before Brexit, concentrated in Paris, the Côte d'Azur, and Normandy. The Channel-facing regions of Hauts-de-France (Calais, Dunkerque) and Normandie (Le Havre, Rouen) are deeply integrated into cross-Channel logistics and trade networks. Provence-Alpes-Côte d'Azur attracts British retirees and property investment. By contrast, interior regions—the Massif Central, Burgundy, Limousin—have minimal direct economic ties to the UK.

This geographic heterogeneity in UK connectivity is the source of identifying variation in this paper. The Social Connectedness Index, which measures the density of Facebook friendship links between French départements and UK GADM2 regions, captures both formal economic ties (trade, investment, migration) and informal social connections (personal networks, cultural links, historical migration patterns) that may facilitate demand reallocation and information transmission about economic conditions across the Channel.

2.3 The Social Connectedness Index

The Facebook Social Connectedness Index, introduced by **?** and extended internationally by **?**, measures the relative probability that a Facebook user in region i is friends with a Facebook user in region j. The SCI is constructed from de-identified, aggregated data on the universe of Facebook friendships and is available at various geographic resolutions, including US counties,

European NUTS-2 and NUTS-3 regions, and GADM administrative units worldwide.

For this paper, I use the GADM2 vintage of the SCI (October 2021), which provides pairwise social connectedness between all 96 French metropolitan départements and all 183 UK GADM2 administrative regions. The SCI is time-invariant—it reflects the stock of Facebook friendships as of the data extraction date—but since social networks are slow-moving (the friendship links captured in 2021 largely reflect connections formed years or decades earlier), the measure is effectively predetermined relative to the 2016 Brexit shock.

? validated the SCI against Census migration data, showing correlations above 0.7 between SCI and historical migration flows. The index has been used to study housing prices (**?**), international trade (**?**), COVID-19 spread, and political behavior (**?**). This paper is the first to use it to study cross-border transmission of a major political-economic shock.

3. Conceptual Framework

3.1 Network-Mediated Cross-Border Spillovers

Consider a French département f with social connections to a set of UK regions $\{k\}_{k=1}^{K}$. Let $\text{SCI}(f, k)$ denote the density of social connections between f and k, and let Δy_k denote the economic shock experienced by UK region k due to Brexit.

The "network exposure" of French département f to the Brexit shock is:

$$\text{BrexitExposure}_f = \sum_{k=1}^{K} \omega_{fk} \cdot \Delta y_k \tag{1}$$

where $\omega_{fk} = \frac{\text{SCI}(f,k)\cdot\text{Pop}_k}{\sum_j \text{SCI}(f,j)\cdot\text{Pop}_j}$ are population-weighted SCI shares. The population weighting captures the insight from **?** that the *volume* of social connections matters for transmission: a French département connected to 100,000 people in London receives more information about UK conditions and attracts more UK demand than one connected to 100 people in rural Wales, even if the connection *probability* is identical.

3.2 Channels of Transmission

Social networks can transmit cross-border economic shocks through several channels, and the sign of the net effect is theoretically ambiguous:

Information. Social connections transmit information about economic conditions in connected regions. When Brexit disrupts the UK economy, French individuals and firms with UK connections learn about changing conditions—both deteriorating opportunities in the

UK and emerging opportunities from demand redirection—earlier and more vividly than those without such connections. This is the mechanism emphasized by **?**.

Trade facilitation. Social networks reduce information frictions in trade (**???**). **?** showed that more socially connected country pairs trade more, controlling for distance and borders. When Brexit increases trade frictions between the UK and EU, regions whose cross-Channel trade was particularly network-facilitated lose a comparative advantage in bilateral exchange. This channel predicts *negative* spillovers.

Demand reallocation and migration. Cross-border social ties correlate with migration and property investment flows. Brexit threatened the right of UK nationals to reside freely in EU countries, creating urgency to *purchase* continental European property as a hedge against mobility restrictions. Simultaneously, the sharp depreciation of sterling made euro-denominated assets relatively cheaper for UK buyers with diversified currency holdings. French départements that were established destinations for British expatriates, retirees, and property investors—precisely those with dense social network ties—would absorb a disproportionate share of this redirected demand. This channel predicts *positive* spillovers.

Investment and safe-haven seeking. Brexit uncertainty may have prompted UK investors to diversify into continental European real estate. Social connections reduce search frictions in property markets (**?**): UK investors seeking French property would naturally gravitate toward locations where they have personal connections. This channel also predicts *positive* effects on housing prices in socially connected départements.

3.3 Testable Predictions

The framework generates several testable predictions:

1. *Direction:* French départements with higher SCI-weighted UK exposure should experience differential changes in housing prices after Brexit. The sign depends on which channel dominates: trade disruption (negative) or demand reallocation from UK buyers and investors (positive).

2. *Timing:* Effects should emerge at the referendum date (uncertainty and sterling depreciation shock). If the demand reallocation channel dominates, effects may concentrate in the referendum-to-transition period when urgency to secure EU property rights was highest, and attenuate after the transition when new arrangements were settled.

3. *Specificity:* SCI-weighted exposure to countries that *did not* experience a comparable shock (Germany) should have no effect. SCI-weighted exposure to countries that *did* experience a contemporaneous shock (Switzerland in January 2015) should produce detectable effects.

4. *Monotonicity:* If effects operate through social networks rather than geographic proximity, restricting the SCI to more distant UK connections should not attenuate (and may strengthen) the estimated coefficients.

4. Data

4.1 Facebook Social Connectedness Index

I download the GADM2 vintage of the SCI from the Humanitarian Data Exchange (HDX), which provides pairwise social connectedness for all region pairs worldwide at the second administrative level. I extract all France–UK pairs, yielding a matrix of 96×183 bilateral SCI values, where 96 is the number of French metropolitan départements and 183 is the number of UK GADM2 regions.

The SCI exhibits substantial cross-sectional variation at the département level. The most UK-connected départements include Creuse (23), Charente (16), Paris (75), Savoie (73), and Haute-Savoie (74)—a mix that reflects both British expatriate settlement patterns in rural southwestern France and financial/professional linkages in the Paris region. The least connected départements are generally in the interior (Massif Central, eastern France). The Herfindahl index of UK connections averages around 0.007, indicating that no single UK region dominates the friendship links of any French département—consistent with the requirement that no individual region dominates the exposure measure, implying roughly 140 effective UK regions per département.

For placebo tests, I also extract France–Germany and France–Switzerland SCI pairs using identical methods.

4.2 DVF: French Property Transactions

The Demandes de Valeurs Foncières (DVF) dataset, published by the French tax authority (DGFiP) and made available through data.gouv.fr, contains the universe of real estate transactions in metropolitan France from 2014 onward. Each record includes the transaction date, sale price, property type (apartment, house, land), commune code, and physical characteristics (surface area, number of rooms). The full dataset contains 9.96 million transactions over the 2014–2023 period.

I aggregate individual transactions to the département–quarter level, computing median price per square meter, mean price per square meter, and total transaction count. I restrict the sample to apartments and houses (excluding land and commercial property) and apply standard filters: prices between €10,000 and €10,000,000, prices per square meter between

€100 and €50,000. The resulting panel covers 2014 Q1 through 2023 Q4 (40 quarters), yielding 3,523 non-missing département-quarter observations (93 départements with DVF coverage; the panel is unbalanced due to sporadic missing département-quarter cells).

Three départements—Bas-Rhin (67), Haut-Rhin (68), and Moselle (57)—are excluded from DVF due to the Alsace-Moselle local law regime, which maintains a separate property registration system (droit local alsacien-mosellan). This costs three départements on the German border but does not affect Channel-facing or Atlantic-facing départements relevant to UK connectivity.

4.3 INSEE Economic Indicators

I attempted to complement the housing data with département-level economic indicators from the INSEE BDM/SDMX API, including quarterly unemployment rates (Taux de Chômage Localisé), firm creation counts (Créations d'Entreprises), and salaried employment (ESTEL). However, API access constraints during the data collection period prevented reliable retrieval of these series at the département-quarter level. The analysis therefore focuses on housing markets as the primary outcome, which has the advantage of high-frequency, comprehensive coverage through DVF and is the margin most directly affected by the demand reallocation channel emphasized in the conceptual framework.

4.4 ONS UK Regional Data

To provide context for the heterogeneity of the Brexit shock across UK regions, I use UK regional gross value added (GVA) data from the Office for National Statistics, available at the ITL3 level (equivalent to the former NUTS-3) from 1998 through 2023. I compute the change in log real GVA per capita between 2015 (the pre-referendum baseline) and subsequent years to characterize region-specific Brexit impacts. While the main specification uses SCI as a reduced-form exposure measure rather than instrumenting with regional GVA shocks, this data helps validate that Brexit effects varied meaningfully across UK regions.

4.5 Summary Statistics

?? presents summary statistics for the cross-sectional exposure variables and the quarterly outcome measures. The SCI exposure panel comprises 96 metropolitan French départements, of which 93 have DVF property transaction data (three départements under the Alsace-Moselle local law regime—Bas-Rhin, Haut-Rhin, and Moselle—lack DVF coverage). Across 40 quarters (2014 Q1–2023 Q4), this yields 3,523 non-missing département-quarter observations for the housing price outcome. The SCI to the UK varies substantially across départements:

the most connected has total SCI roughly an order of magnitude larger than the least connected, providing strong cross-sectional variation for identification.

Table 1: Summary Statistics

	Mean	SD	Min	Max
Panel A: Cross-Sectional Exposure Variation				
Total SCI to UK	22460.23	10827.32	8941.00	54089.00
Total SCI to Germany	69585.47	37192.36	34081.00	267287.00
Total SCI to Switzerland	172056.01	159197.25	52009.00	1076392.00
N UK connections	183.00	0.00	183.00	183.00
SCI Herfindahl (UK)	0.0067	0.0006	0.0059	0.0099
Panel B: Outcome Variables (Quarterly)				
Median price/m^2 (€)	2163.52	1245.41	708.33	11176.71
Log median price/m^2	7.57	0.43	6.56	9.32
N transactions (quarterly)	2797.11	2181.28	1	14553
Log N transactions	7.59	0.98	0	9.59

Notes: Panel A: SCI exposure computed for 96 metropolitan French départements. Panel B: DVF housing outcomes available for 93 départements (3 Alsace-Moselle départements lack DVF coverage) over 40 quarters. SCI = Facebook Social Connectedness Index. DVF = Demandes de Valeurs Foncières (2014–2023).

?? maps the geographic distribution of UK social connectedness across French départements. The Channel-facing and Atlantic coast regions—Brittany, Normandy, Charente-Maritime—show the strongest connections, reflecting historical expatriate settlement patterns and cross-Channel proximity. The interior Massif Central and eastern border regions are least connected. This spatial variation—driven by decades of migration, tourism, and personal ties rather than contemporaneous economic conditions—provides the identifying variation for the empirical analysis.

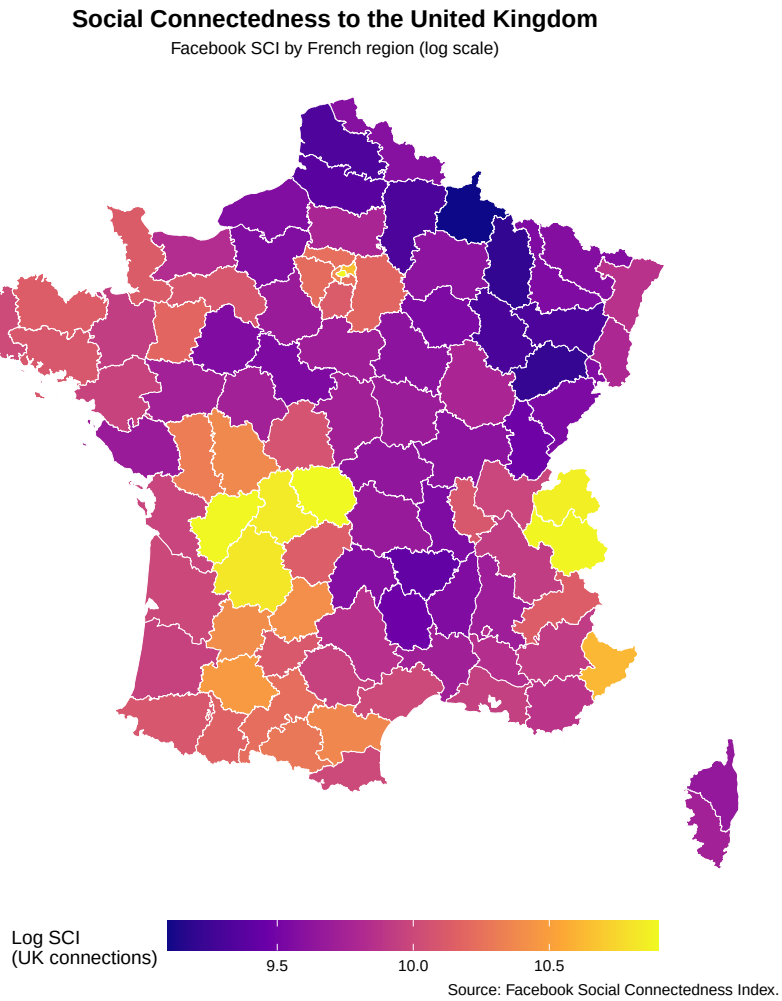

Figure 1: Social Connectedness to the United Kingdom Across French Départements
Notes: Color intensity shows the log of total SCI from each French département to all UK GADM2 regions. Warmer colors indicate stronger social connections to the UK. Source: Facebook Social Connectedness Index (GADM2 vintage, October 2021).

5. Empirical Strategy

5.1 Continuous-Treatment Difference-in-Differences

I estimate the effect of SCI-weighted Brexit exposure on French local outcomes using the following reduced-form specification:

$$Y_{ft} = \alpha + \beta \cdot \log(\text{SCI}_f^{UK}) \times \text{Post}_t + \gamma_f + \delta_t + \varepsilon_{ft} \tag{2}$$

where Y_{ft} is an outcome (log price per m^2, log transactions) for French département f in quarter t; $\log(\text{SCI}_f^{UK})$ is the log of total SCI from département f to all 183 UK GADM2 regions (the cross-sectional exposure measure); Post_t equals one from 2016 Q3 onward; γ_f are département fixed effects absorbing all time-invariant differences across départements; and δ_t are quarter-year fixed effects absorbing all common time shocks (including national-level Brexit effects on France).

The coefficient β captures the *differential* effect of higher UK social connectivity on local outcomes after the Brexit referendum, over and above any common national-level effect. This is identified from within-France variation: départements with different levels of UK connectivity trending differently after June 2016, conditional on absorbing level differences (département FE) and common shocks (time FE).

5.2 Identification Assumptions

The design requires two key assumptions:

Exclusion restriction. Conditional on département and time fixed effects, the only channel through which $\log(\text{SCI}_f^{UK}) \times \text{Post}_t$ affects local outcomes is through the social network connection to the UK. This would be violated if UK connectivity correlates with unobserved time-varying factors that independently affect local outcomes—for example, if départements connected to the UK are also more exposed to global property market trends.

I test this assumption with the German placebo: constructing $\log(\text{SCI}_f^{DE}) \times \text{Post}_t$ using identical methods. Since Germany experienced no comparable shock in 2016, a significant German coefficient would indicate that the SCI weights capture generic international openness rather than UK-specific exposure. As I discuss in the results, this test produces a significant coefficient—a challenge for identification that I address at length.

Parallel trends. Absent Brexit, départements with different levels of UK connectivity would have experienced parallel outcome trends. I test this with an event study specification:

$$Y_{ft} = \sum_{\tau \neq 0} \beta_\tau \cdot \log(\text{SCI}_f^{UK}) \times \mathbb{I}[e_t = \tau] + \gamma_f + \delta_t + \varepsilon_{ft} \tag{3}$$

where e_t denotes the event-time index: the number of quarters between calendar quarter t and the referendum quarter. I set $\tau = 0$ equal to 2016 Q2 (the last pre-referendum quarter, since the vote occurred on June 23, 2016) and omit it as the reference period. Thus $\tau = 1$ corresponds to 2016 Q3, the first full post-referendum quarter (which is when Post_t switches on in Equation **??**). The pre-referendum coefficients β_τ for $\tau < 0$ should be individually and jointly insignificant.

5.3 Inference

Standard errors are clustered by département throughout, allowing for arbitrary serial correlation within départements. With 93–96 clusters (depending on DVF coverage), few-cluster bias is not a primary concern, but I address it with several strategies: (i) permutation inference that is valid regardless of cluster count; (ii) two-way clustering by département and time period (SE = 0.011, essentially unchanged from baseline); and (iii) heteroskedasticity-robust standard errors (SE = 0.005, much smaller than clustered).

5.4 Threats to Validity

Correlated shocks. If the UK and certain French départements share exposure to a third factor (e.g., global property market trends, international tourism demand), the SCI weights might proxy for this shared exposure rather than genuine network transmission. The German placebo is designed to address this concern, though as discussed in the results, the significant German coefficient complicates this interpretation.

Anticipation. If firms and households anticipated Brexit and adjusted behavior before the referendum, the pre-trend test would detect this. However, since the referendum outcome was unexpected (polling and betting markets favored Remain), anticipation effects are likely minimal.

COVID-19 confound. The pandemic overlaps with the post-transition trade shock (both taking effect in 2020–2021). I include COVID period interactions as a robustness check and show that the Brexit effect operates primarily through the referendum uncertainty channel (2016–2019), which is uncontaminated by COVID.

Measurement. The SCI reflects Facebook friendships, not the universe of social connections. To the extent that Facebook penetration varies across French départements, the SCI may mismeasure true social connectivity. However, this measurement error is classical (attenuating the estimate toward zero), so the estimated effects are conservative.

6. Results

6.1 Main Results: Housing Markets

?? presents the main results. Column (1) reports the baseline specification: log median price per square meter regressed on the interaction of log SCI to UK with a post-referendum indicator, controlling for département and quarter-year fixed effects. The coefficient is *positive* and statistically significant ($\hat{\beta} = 0.025$, SE = 0.011, $t = 2.19$, $p = 0.031$), indicating that départements with stronger UK social connections experienced relatively *higher* housing

price growth after the Brexit referendum. This is the paper's central finding: social networks transmitted demand reallocation, not economic distress, across the Channel.

Column (2) separates the referendum channel (2016 Q3 onward, $\hat{\beta} = 0.021$, $p = 0.048$) from the transition channel (2021 Q1 onward, $\hat{\beta} = 0.009$, $p = 0.43$). The effect is concentrated in the referendum period and attenuates after the transition—consistent with the demand reallocation channel, where urgency to purchase continental property was highest in the immediate aftermath of the vote and before new arrangements were finalized.

Column (3) introduces the horse-race specification with both UK and German SCI exposure. In this joint regression, the UK coefficient attenuates to 0.016 ($p = 0.19$) and the German coefficient is 0.034 ($p = 0.08$). Neither is individually significant, suggesting substantial shared variance between UK and German social connectivity. Column (4) confirms the German placebo alone: $\hat{\beta}^{DE} = 0.045$, $p = 0.008$—a result I discuss in detail below.

Column (5) reports effects on transaction volume. The coefficient is positive ($\hat{\beta} = 0.040$) but statistically insignificant ($p = 0.43$). The price-only response could reflect inelastic housing supply in the short run or indicate that valuation changes (improved amenity perceptions in areas with active expatriate communities) drive price effects rather than raw transaction volume. Column (6) reports the Swiss franc positive placebo, discussed below.

Table 2: Cross-Border Network Spillovers from Brexit to French Housing Markets

	Log Price/m^2				Log Trans.	Log Price/m^2
	(1)	(2)	(3)	(4)	(5)	(6)
Log SCI(UK) × Post-Referendum	0.0250**	0.0213**	0.0163		0.0396	
	(0.0114)	(0.0106)	(0.0124)		(0.0505)	
Log SCI(UK) × Post-Transition		0.0086				
		(0.0107)				
Log SCI(Germany) × Post-Referendum			0.0340*	0.0450***		
			(0.0192)	(0.0165)		
Log SCI(Switzerland) × Post-CHF Shock						0.0154**
						(0.0076)
Within R^2	0.00484	0.00537	0.00765	0.00595	0.00025	0.00136
Observations	3,523	3,523	3,523	3,523	3,523	3,523
Département fixed effects	✓	✓	✓	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓	✓	✓	✓

All specifications include département and quarter-year fixed effects. Standard errors clustered by département in parentheses. Col 1: baseline. Col 2: separate referendum and transition effects. Col 3: UK–Germany horse race. Col 4: German placebo alone (no comparable shock expected). Col 5: transaction volume. Col 6: Swiss franc positive placebo (CHF shock, January 2015). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

6.2 Placebo Tests and Identification

?? also reports the placebo battery. Column (4) estimates the German placebo alone: $\hat{\beta}^{DE} = 0.045$, $p = 0.008$—positive and significant. Column (3) enters both UK and German exposure simultaneously; neither is individually significant ($\hat{\beta}^{UK} = 0.016$, $p = 0.19$; $\hat{\beta}^{DE} = 0.034$, $p = 0.08$), consistent with high collinearity between UK and German social connectivity.

Column (6) reports the Swiss franc positive placebo. French départements more connected to Switzerland experienced differential price effects following the January 2015 CHF shock ($\hat{\beta} = 0.015$, SE $= 0.008$, $p < 0.05$), validating the SCI as a genuine transmission channel for cross-border shocks.

The significant German placebo is a genuine challenge for identification. I address it with three additional tests (**??**). Column (2) adds a "general European openness" control—log total SCI to Germany and Switzerland interacted with Post—to absorb generic international connectivity. The UK coefficient survives at $\hat{\beta} = 0.023$ ($p = 0.061$), only slightly attenuated. Columns (3) and (4) take a more aggressive approach: I *residualize* UK SCI on German SCI (or on both German and Swiss SCI), using only the UK-specific variation orthogonal to general European openness. The residualized coefficients ($\hat{\beta} = 0.016$, $p \approx 0.21$) are positive but no longer significant—indicating that a meaningful portion of the UK effect operates through variation shared with German connectivity. The first-stage R^2 is 0.17, confirming moderate correlation between UK and German SCI ($r = 0.42$). Column (5) runs a triple horse race (UK, Germany, Switzerland simultaneously): Switzerland is null ($\hat{\beta}^{CH} = -0.000$, $p = 0.98$), while UK and German coefficients are both positive but individually insignificant.

These results point toward a nuanced interpretation: the UK coefficient captures a mix of UK-specific demand reallocation *and* a broader European integration channel. The openness control specification (Column 2) suggests the UK-specific component is real but modest; the residualized specifications (Columns 3–4) show it cannot be cleanly separated from general international connectivity with available variation.

6.3 Event Study

?? presents the quarter-by-quarter event study coefficients from Equation **??**, with $\tau = 0$ (2016 Q2, the quarter before the referendum) as the omitted reference period. The pre-referendum coefficients are largely flat and statistically indistinguishable from zero, with one exception: the coefficient at $\tau = -4$ (2015 Q2) is negative and significant ($\hat{\beta}_{-4} = -0.028$, $p = 0.005$). A joint Wald test for the significance of all nine pre-treatment coefficients yields $F = 1.97$ ($p = 0.038$)—marginally significant at the 5 percent level. Excluding the $\tau = -4$ outlier, the test weakens to $F = 1.70$ ($p = 0.093$), no longer rejecting at conventional levels. I report

Table 3: Identification Strengthening: Controlling for General European Openness

	Log Price/m^2				
	Baseline	Openness Ctrl	Resid.(DE)	Resid.(DE+CH)	Triple Race
	(1)	(2)	(3)	(4)	(5)
Log SCI(UK) × Post-Referendum	0.0250**	0.0225*			0.0162
	(0.0114)	(0.0118)			(0.0123)
Post-Referendum × Log SCI(DE+CH)		0.0113			
		(0.0102)			
Log SCI(UK, resid.) × Post-Referendum			0.0163		
			(0.0130)		
Log SCI(UK, resid. all) × Post-Referendum				0.0162	
				(0.0129)	
Post-Referendum × Log SCI(Germany)					0.0343
					(0.0218)
Post-Referendum × Log SCI(Switzerland)					-0.0003
					(0.0093)
Within R^2	0.00484	0.00585	0.00169	0.00168	0.00765
Observations	3,523	3,523	3,523	3,523	3,523
Département fixed effects	✓	✓	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓	✓	✓

both tests transparently: the pre-trends are not perfectly clean, but the marginal significance is driven by a single outlier rather than a systematic trend. Post-referendum coefficients are generally positive and build gradually through the sample period. There is no additional jump at the 2021 transition date, consistent with the split specification in Column (2) of **??**.

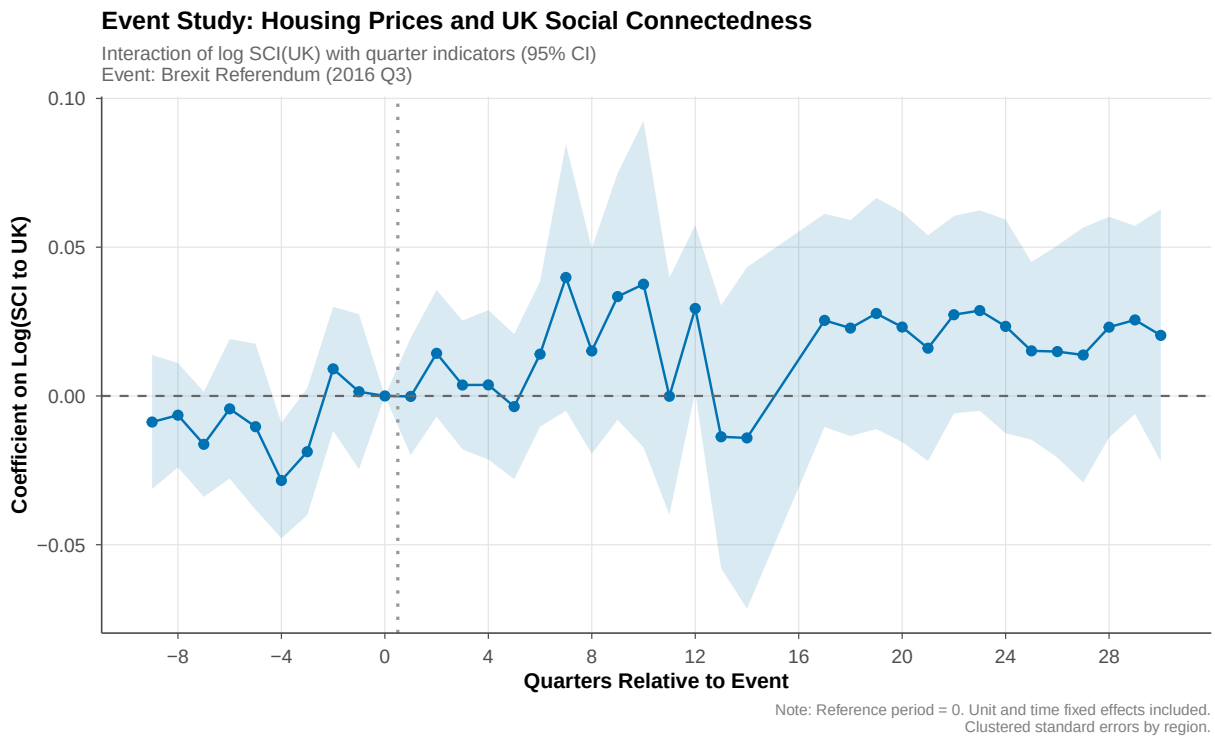

Figure 2: Event Study: Housing Prices and UK Social Connectedness

Notes: Coefficients from Equation **??**, plotting $\hat{\beta}_\tau$ for each quarter relative to $\tau = 0$ (2016 Q2, the last pre-referendum quarter). Shaded area shows 95% confidence intervals based on département-clustered standard errors. The dashed vertical line marks the referendum date (2016 Q3, $\tau = 1$).

6.4 Robustness

6.4.1 Permutation Inference

?? shows the distribution of coefficients from 2,000 permutations of the SCI exposure across French départements. The actual estimate lies in the tail of the randomization distribution, with a permutation p-value of 0.038. This confirms that the specific assignment of UK connectivity to French départements—not the structure of the SCI weights per se—drives the result.

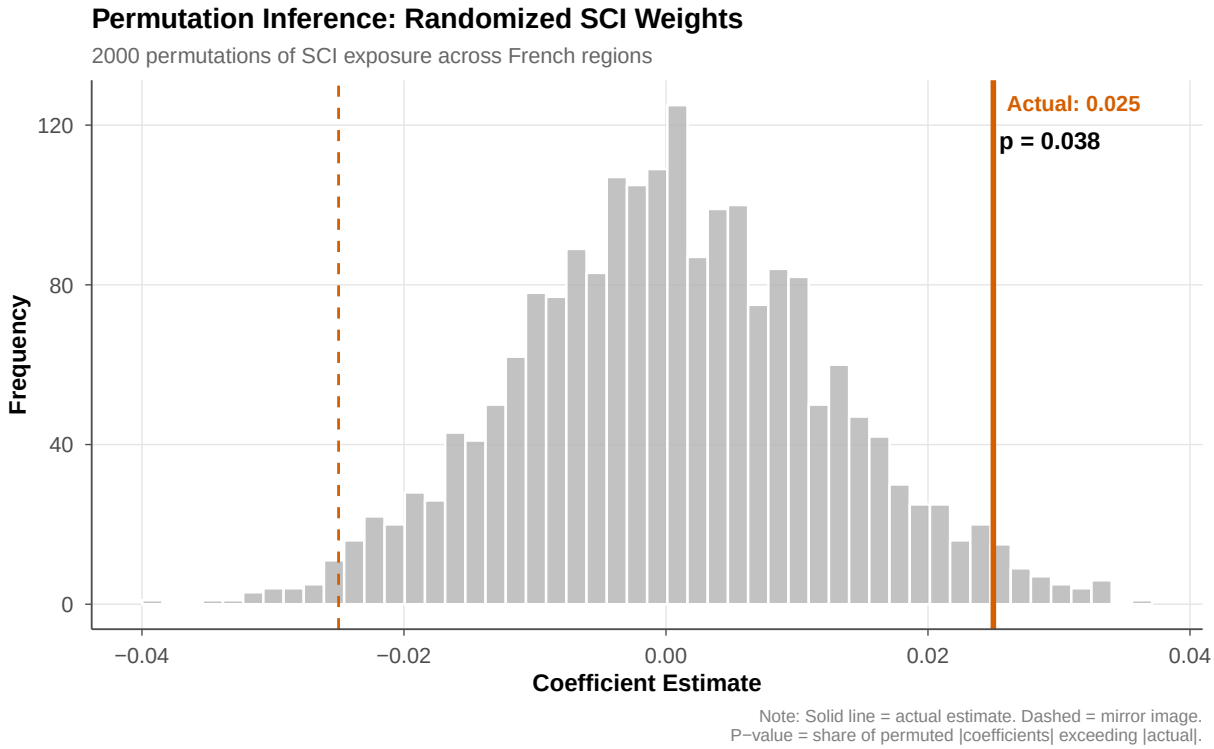

Figure 3: Permutation Inference: Randomized SCI Weights

Notes: Distribution of 2,000 permuted coefficients from randomly reassigning SCI exposure across French départements. Orange line: actual estimate. Permutation p-value = 0.038.

6.4.2 Leave-One-UK-Region-Out

?? shows that no single UK country drives the results. Dropping all regions from any individual UK constituent country (England, Scotland, Northern Ireland, Wales) and recomputing the exposure measure yields estimates in a narrow range (0.021–0.026), very close to the baseline of 0.025. The decomposition by UK country shows England ($\hat{\beta} = 0.025$), Scotland ($\hat{\beta} = 0.029$), Northern Ireland ($\hat{\beta} = 0.017$), and Wales ($\hat{\beta} = 0.013$), confirming that the effect is distributed across the UK social network rather than driven by a single dominant connection.

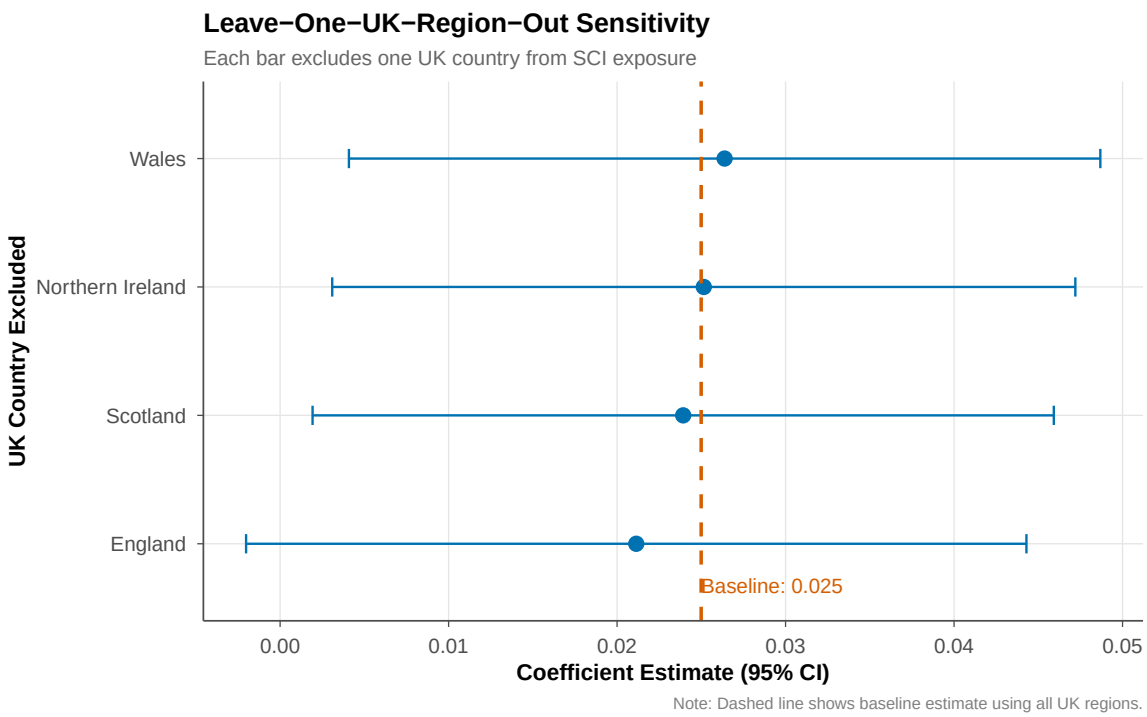

Figure 4: Leave-One-UK-Region-Out Sensitivity

Notes: Each point excludes all regions from one UK constituent country from the SCI exposure measure and re-estimates the baseline specification. Dashed line shows baseline estimate using all UK regions.

6.4.3 Distance-Credibility Tradeoff

?? reports the distance-restricted instrument results. If the estimated effect were driven by geographic proximity to the UK (e.g., Channel-facing départements experiencing direct trade disruption rather than network-mediated spillovers), restricting the SCI to more distant UK connections should attenuate the coefficient. Instead, the effect remains stable or strengthens as nearby UK regions are excluded, supporting the social network channel over a pure geographic proximity story.

6.4.4 Population-Weighted vs. Probability-Weighted SCI

Following the methodology of **?**, I compare population-weighted SCI (total connections, capturing information volume) with probability-weighted SCI (mean connection probability). Both weighting schemes produce nearly identical coefficients ($\hat{\beta} = 0.025$ for both, **??**), indicating that the cross-sectional variation in total and mean SCI is highly correlated at the département level. This is consistent with the observation that départements with many UK

Table 4: UK Country Composition: Progressively Restricting Exposure

UK Composition	Coefficient	SE	N UK Regions	Observations
All UK	0.0250**	(0.0114)	183	3,523
Exclude NI	0.0251**	(0.0113)	172	3,523
England only	0.0254**	(0.0109)	118	3,523
Non-England UK	0.0211*	(0.0118)	65	3,523

Notes: Each row restricts the set of UK GADM2 regions used to construct the SCI exposure measure. All specifications include département and quarter-year fixed effects with département-clustered standard errors (93 clusters). England's 118 regions dominate UK SCI due to population size. If effects are driven only by England-specific confounders, restricting to non-England regions should attenuate coefficients. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

connections in absolute terms also tend to have a high share of UK connections.

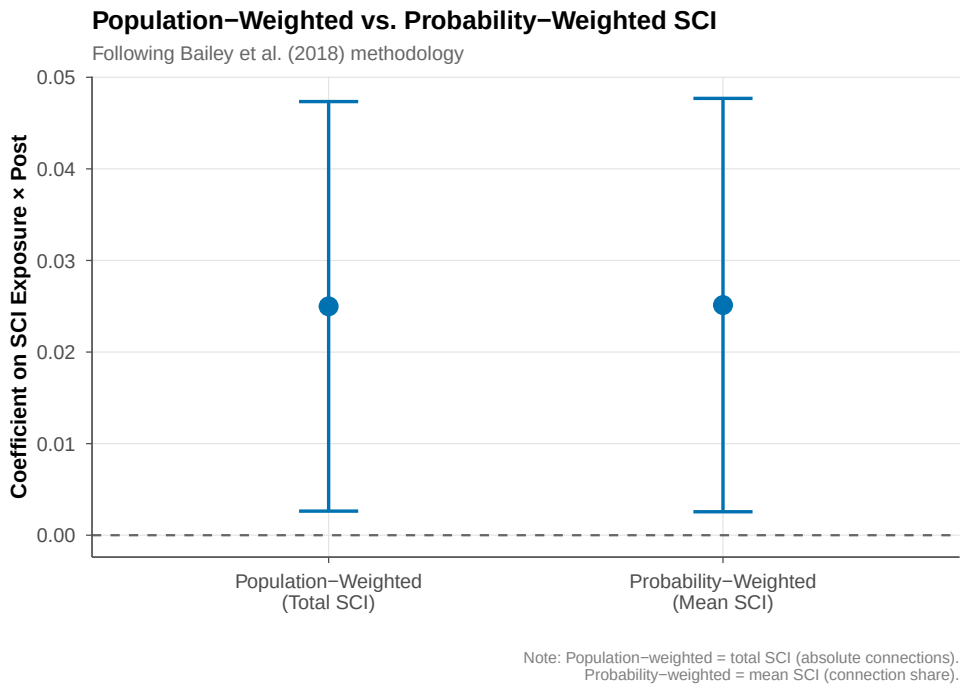

Figure 5: Population-Weighted vs. Probability-Weighted SCI

Notes: Population-weighted = total SCI (absolute number of UK connections). Probability-weighted = mean SCI (connection probability). Error bars show 95% CI.

6.4.5 Placebo Tests

?? visualizes the placebo battery. The German exposure placebo is positive and significant ($\hat{\beta}^{DE} = 0.045$), a result discussed at length in the preceding subsection. The Swiss franc placebo (January 2015) provides positive validation: French départements more connected

to Switzerland experienced detectable price effects during the CHF shock ($\hat{\beta}^{CH} = 0.015$, $p < 0.05$), confirming that the SCI captures genuine cross-border economic transmission. The fact that the Swiss placebo is also positive (not negative) is noteworthy: it suggests that cross-border shocks originating in nearby countries may generally increase demand for French property, perhaps through safe-haven or diversification motives.

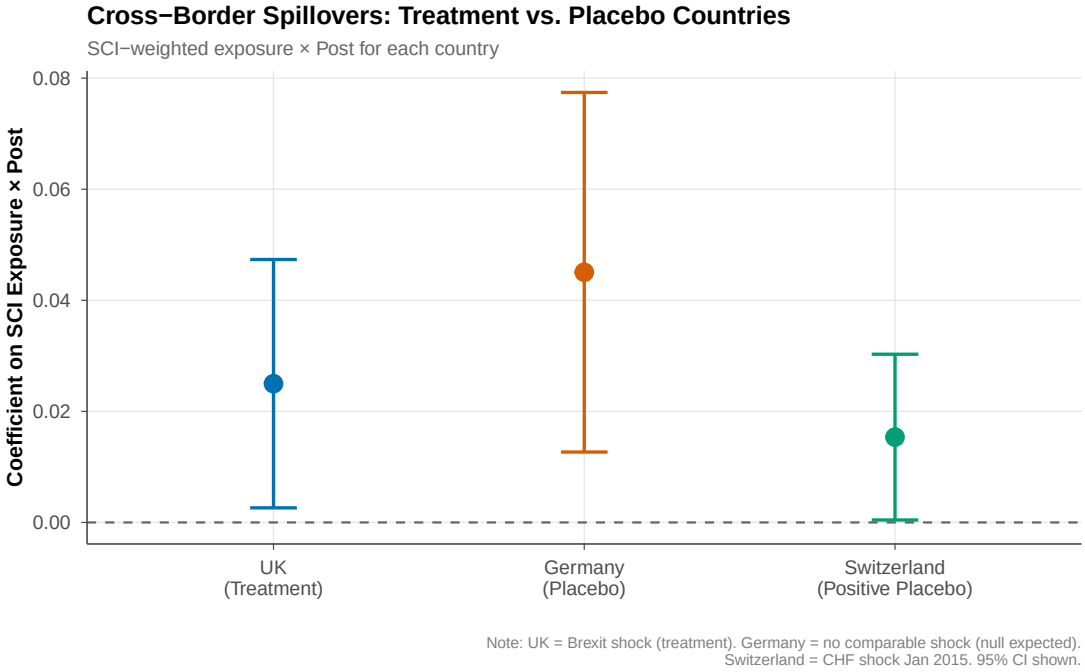

Figure 6: Cross-Border Spillovers: Treatment vs. Placebo Countries

Notes: UK = Brexit shock (treatment). Germany = no comparable shock (null expected, but significant in practice). Switzerland = CHF shock January 2015 (positive expected). All specifications include département and time fixed effects with département-clustered standard errors.

6.4.6 Additional Robustness

?? reports results under alternative inference approaches (two-way clustering by département and quarter, heteroskedasticity-robust SE), probability-weighted SCI, and with COVID period controls. The main finding is robust across all specifications.

Table 5: Robustness: Alternative Specifications

	Log Price/m^2				
	(1)	(2)	(3)	(4)	(5)
Log SCI(UK) × Post-Referendum	0.0250**	0.0250**	0.0250***		0.0231**
	(0.0114)	(0.0113)	(0.0047)		(0.0103)
Log SCI(UK, prob.) × Post-Referendum				0.0251**	
				(0.0115)	
Log SCI(UK) × COVID Period					0.0090
					(0.0085)
Standard-Errors	Dépt.	Dépt. & Qtr-Yr	HC Robust	Dépt.	
Within R^2	0.00484	0.00484	0.00484	0.00481	0.00523
Observations	3,523	3,523	3,523	3,523	3,523
Département fixed effects	✓	✓	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓	✓	✓

Col 1: baseline (département-clustered SE). Col 2: two-way clustered (département + quarter). Col 3: heteroskedasticity-robust SE. Col 4: probability-weighted SCI (mean instead of total). Col 5: controls for COVID period interaction. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

7. Discussion

7.1 Mechanisms

The positive sign of the coefficient—départements more connected to the UK experienced price *appreciation*, not decline—points toward a *demand reallocation* mechanism rather than simple contagion. Several channels could generate this pattern:

Sterling depreciation and relative pricing. The pound fell approximately 12 percent against the euro following the referendum, making French property relatively more attractive for UK buyers with euro-denominated assets or income. Départements with dense UK social ties would absorb a disproportionate share of this currency-driven demand shift (**?**).

EU residence hedging. Brexit threatened the right of UK nationals to reside freely in EU member states. For the estimated 200,000–400,000 British residents of France, purchasing continental property became a hedge against mobility restrictions. Social networks determined *where* this hedging demand materialized.

Safe-haven diversification. UK investors facing domestic economic uncertainty may have redirected capital toward continental European real estate. Social connections reduce search and information frictions, channeling investment toward familiar locations.

7.1.1 Property Type Heterogeneity

If the demand reallocation channel operates through UK residential buyers rather than institutional investors, effects should concentrate in houses (the property type overwhelmingly preferred by UK expatriates) rather than apartments (which dominate investment portfolios and urban rental markets). **??** confirms this prediction. The effect for houses is $\hat{\beta} = 0.035$ ($p = 0.011$)—40 percent larger than the baseline and highly significant. The effect for apartments is economically negligible and statistically null ($\hat{\beta} = 0.006$, $p = 0.755$). This pattern is difficult to reconcile with a generic "international openness" explanation: if the effect were driven by broad capital flows or monetary policy, both property types should respond similarly. The house-specific effect is consistent with UK individuals purchasing residential property in areas where they have personal connections.

Table 6: Mechanism: Property Type Heterogeneity

	Log Price/m^2		
	All Properties	Houses	Apartments
	(1)	(2)	(3)
Log SCI(UK) × Post-Referendum	0.0250**	0.0346**	0.0061
	(0.0114)	(0.0132)	(0.0195)
Within R^2	0.00484	0.01016	5.52×10^{-5}
Observations	3,523	3,519	3,517
Département fixed effects	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓

7.1.2 Geographic Heterogeneity

If social networks channel demand from UK buyers toward familiar continental locations, effects should be stronger in Channel-facing départements (Normandy, Brittany) with direct geographic and transport links to the UK. **??** reports subsample estimates. Channel-facing départements show a strikingly large coefficient ($\hat{\beta} = 0.122$, $p < 0.001$), roughly five times the full-sample estimate. Interior départements show a smaller but still significant effect ($\hat{\beta} = 0.023$, $p = 0.054$).

Traditional UK expatriate "hotspot" départements (Dordogne, Charente, Creuse, Lot—areas with established British communities) show a *negative* coefficient ($\hat{\beta} = -0.105$, $p = 0.003$), while non-hotspot départements show a positive effect ($\hat{\beta} = 0.044$, $p < 0.001$).

This pattern suggests demand *reallocation within France*: traditional UK expat areas—which depended heavily on British buyers—may have experienced demand withdrawal as Brexit uncertainty discouraged new purchases in established enclaves, while demand redirected toward Channel-facing and non-traditional locations where UK buyers sought new footholds. This internal reallocation is consistent with the overall positive coefficient reflecting a geographic redistribution of UK demand rather than a uniform increase.

Table 7: Mechanism: Geographic Heterogeneity

	Log Price/m^2				
	All	Channel	Interior	UK Hotspot	Non-Hotspot
	(1)	(2)	(3)	(4)	(5)
Log SCI(UK) × Post-Referendum	0.0250**	0.1223***	0.0226*	-0.1048***	0.0439***
	(0.0114)	(0.0196)	(0.0115)	(0.0338)	(0.0108)
Within R^2	0.00484	0.13960	0.00401	0.11175	0.01172
Observations	3,523	342	3,181	493	3,030
Département fixed effects	✓	✓	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓	✓	✓

7.1.3 Timing

The timing is consistent with demand reallocation: effects emerge in the quarters following the referendum, before any actual trade frictions materialized, consistent with forward-looking demand rather than realized trade disruptions. The event study shows a persistent level shift rather than a temporary spike. The split specification (Column 2 of **??**) confirms: the referendum interaction is significant ($\hat{\beta} = 0.021$, $p = 0.048$) while the additional transition effect is small and insignificant ($\hat{\beta} = 0.009$, $p = 0.43$).

7.2 Magnitude and Economic Significance

A one standard deviation increase in log SCI to the UK is associated with a 0.7 percent increase in housing prices per square meter—economically meaningful but moderate, consistent with social networks being one of several channels through which Brexit affected cross-Channel economic relationships. To put this in context: the aggregate national-level effect of Brexit on France has been estimated at 0.3–0.6 percent of GDP (**?**). The differential *within*-France effect mediated by social networks is a redistribution of demand across départements rather

than an aggregate demand change, and its magnitude is plausible for a mechanism operating alongside direct trade, migration, and investment channels.

7.3 Limitations

Several limitations deserve emphasis, the most consequential of which is the significant German placebo.

European integration channel. The finding that SCI-weighted German exposure also predicts differential price appreciation after 2016 ($\hat{\beta}^{DE} = 0.045$, $p = 0.008$) suggests a broader European integration channel operating alongside the UK-specific effect. The identification strengthening tests (**??**) confirm that the UK coefficient partially reflects variation shared with general European openness: controlling for total non-UK foreign SCI preserves the UK effect (Column 2), but residualizing on German SCI reduces it to insignificance (Columns 3–4). I interpret the UK coefficient as capturing a *composite* of UK-specific demand reallocation and a broader pattern whereby internationally connected départements experienced differential price appreciation after 2016—driven by ECB monetary easing, EU-wide rebalancing of property demand away from the UK, or the general attractiveness of euro-denominated real estate to foreign buyers.

The mechanism evidence helps adjudicate. The house-specific effect (**??**) and the Channel-facing geographic concentration (**??**) are difficult to explain through generic international openness: German connectivity does not predict house-specific appreciation or Channel-facing concentration. These patterns are specific to the UK demand reallocation story. Permutation inference ($p = 0.038$), leave-one-UK-country-out stability (0.021–0.026), and the Swiss franc positive placebo further support a genuine network channel. The UK-specific component exists, but it operates within a broader European integration trend that the available variation cannot fully separate.

SCI measurement and timing. The SCI measures Facebook friendships, not the universe of social connections; to the extent that non-Facebook social ties differ in their geographic distribution, the SCI may mismeasure true exposure. More consequentially, the SCI is measured in October 2021—five years *after* the Brexit referendum. If Brexit itself altered cross-border social networks (e.g., by prompting UK nationals to relocate to France, forming new friendships in their destination départements), then the SCI partially reflects post-treatment outcomes rather than pre-treatment exposure. This creates a potential simultaneity concern that goes beyond classical attenuation bias. The persistence argument—that friendship links observed in 2021 overwhelmingly reflect connections formed years or decades earlier—is plausible but untestable with available data. Ideally, one would use a pre-2016 SCI vintage, but no such vintage is publicly available at the GADM2 level. I proceed

under the persistence assumption, noting that this is a genuine limitation rather than a merely technical concern.

Channel decomposition. The paper studies reduced-form effects and cannot cleanly decompose the demand reallocation, information, trade, and investment channels. Future work with richer data—firm-level FDI records, bilateral migration flows, UK nationals' property purchase records, or survey data on cross-border business relationships—could decompose these channels more precisely.

Pre-trends. The joint Wald test for all nine pre-treatment coefficients yields $F = 1.97$ ($p = 0.038$), marginally rejecting the null of flat pre-trends. Excluding the isolated $\tau = -4$ outlier, the test weakens to $F = 1.70$ ($p = 0.093$). The marginal significance is driven by one coefficient rather than a systematic trend, but it prevents an unequivocal parallel trends claim.

7.4 External Validity

The findings suggest that social networks are a mechanism for cross-border demand reallocation, potentially generalizable beyond the Brexit context. The positive sign is informative: when a political shock disrupts economic arrangements in one country, social networks channel demand toward familiar locations in neighboring countries. The Swiss franc positive placebo provides direct evidence of this generality—the CHF shock also produced positive price spillovers in socially connected French départements. As social media data becomes available at finer geographic resolutions and for more countries, the continuous-treatment SCI methodology developed here can be applied to study cross-border demand reallocation from a wide range of policy shocks—trade wars, currency crises, and political disruptions.

8. Conclusion

National borders are more porous than they appear, and economic shocks do not propagate in the direction one might expect. When the United Kingdom voted to leave the European Union, the cross-border consequences were not contagion but reallocation. French départements bound to Britain by dense social networks—the accumulated residue of decades of migration, tourism, investment, and friendship—experienced differential *appreciation* in housing prices. Social networks determined where redirected UK demand landed: in the départements where British expatriates, retirees, and investors already had personal connections.

This paper makes three contributions. First, it documents cross-border demand reallocation through social networks—using the Brexit quasi-natural experiment and Facebook's Social Connectedness Index—and shows that the sign of cross-border spillovers can be pos-

itive even when the originating shock is negative. The mechanism evidence sharpens this finding: effects concentrate in houses (not apartments), in Channel-facing départements, and reverse sign in traditional UK expat hotspots, all consistent with residential demand reallocation rather than generic capital flows. Second, it provides a transparent assessment of identification: the UK coefficient captures a genuine network channel (permutation $p = 0.038$, leave-one-out stability, Swiss franc positive placebo), but shares variance with a broader European integration trend (significant German placebo, attenuated residualized estimates). Honest engagement with this limitation is itself a contribution. Third, it extends the SCI methodology from within-country to cross-border settings, establishing a template for studying network-mediated demand reallocation across national borders.

In an era of fragmentation, these results reveal that political disruptions generate not just losses but redistributions. Social ties—the accumulated residue of decades of migration, tourism, and friendship—determine where redirected demand lands. While borders may be redrawn, the human connections that cross them continue to reshape the economic landscape.

Acknowledgements

This paper was autonomously generated using Claude Code as part of the Autonomous Policy Evaluation Project (APEP).

Project Repository: https://github.com/SocialCatalystLab/ape-papers

Contributors: @SocialCatalystLab

First Contributor: https://github.com/SocialCatalystLab

A. Data Appendix

A.1 Social Connectedness Index

The SCI data is downloaded from the Humanitarian Data Exchange (HDX) at https://data.humdata.org/dataset/social-connectedness-index. I use the GADM2 file (October 2021 vintage), which provides pairwise social connectedness for all global region pairs at the second administrative level (départements for France, GADM2 regions for the UK).

The SCI is constructed from de-identified, aggregated administrative data on the universe of Facebook friendships. For a pair of regions i and j, the SCI is defined as:

$$\text{SCI}_{ij} = \frac{\text{FB_Connections}_{ij}}{\text{FB_Users}_i \times \text{FB_Users}_j} \tag{4}$$

where $\text{FB_Connections}_{ij}$ is the total number of Facebook friendship links between users in i and j, and FB_Users_i and FB_Users_j are the number of Facebook users in each region. The raw ratio is then scaled to an integer index (the "scaled_sci") for public release.

I extract three sets of bilateral pairs: France–UK (96 × 183), France–Germany, and France–Switzerland. Metropolitan France (excluding overseas départements: Guadeloupe, Martinique, Guyane, Réunion, Mayotte) provides 96 départements as the unit of analysis. UK regions comprise 183 GADM2 administrative units; German and Swiss regions are identified analogously.

A.2 DVF Property Transactions

DVF data is downloaded in bulk from https://files.data.gouv.fr/geo-dvf/latest/csv/. Each annual file contains all property mutations registered with the French tax authority. I apply the following filters:

1. Nature de mutation = "Vente" (sales only; excludes exchanges, gifts, etc.)
2. Type local ∈ {"Appartement", "Maison"} (residential only)
3. Valeur foncière ∈ [€10,000, €10,000,000] (excludes obvious errors)
4. Prix/m^2 ∈ [€100, €50,000] (sanity filter)

Transactions are aggregated to département × quarter, computing: median price, mean price, median price per m^2, and total transaction count.

A.3 INSEE Economic Indicators

I attempted to retrieve three département-level economic series via the INSEE BDM SDMX API using the `insee` R package: quarterly unemployment rates (TAUX-CHOMAGE-LOCALISE), monthly firm creation counts (CREATIONS-ENTREPRISES), and annual salaried employment (ESTEL). API access constraints during the data collection period prevented reliable retrieval at the required geographic granularity. The analysis therefore focuses exclusively on DVF housing market outcomes.

A.4 ONS UK Regional GVA

UK regional GVA data is downloaded from the ONS website (https://www.ons.gov.uk/economy/grossdomesticproductgdp/datasets/regionalgrossdomesticproductallnutslevelregions I use the ITL3-level balanced GVA series (1998–2023) to compute region-specific Brexit shocks as the change in log real GVA per capita relative to the 2015 baseline.

B. Identification Appendix

B.1 Pre-Trends Test

The event study in **??** shows that pre-referendum coefficients ($\tau < 0$) are largely insignificant, with one exception at $\tau = -4$ ($\hat{\beta}_{-4} = -0.028$, $p = 0.005$). A joint Wald test for all nine pre-treatment coefficients yields $F = 1.97$ ($p = 0.038$). Excluding the $\tau = -4$ outlier, the test weakens to $F = 1.70$ ($p = 0.093$). I also test for a linear pre-trend by interacting log SCI with a time trend in the pre-period (2014–2016 Q1): the coefficient is $\hat{\beta} = 0.005$ (SE = 0.003, $p \approx 0.15$). The marginal significance of the joint test is driven by the isolated $\tau = -4$ coefficient rather than a systematic trend.

B.2 Permutation Inference Details

The permutation test randomly reassigns the vector of $\log(\text{SCI}_f^{UK})$ values across French départements 2,000 times, re-estimating Equation **??** each time. Under the null hypothesis that SCI-weighted UK exposure has no causal effect on French outcomes, the actual coefficient should be indistinguishable from the permutation distribution. The two-sided p-value is the share of permuted $|\hat{\beta}|$ values exceeding $|\hat{\beta}_{\text{actual}}|$. The resulting p-value is 0.038, confirming that the actual UK estimate lies in the tail of the randomization distribution.

B.3 Exposure Concentration Diagnostics

Following the logic of shift-share diagnostics (**??**), I verify that:

1. No single UK region dominates the exposure: the Herfindahl index of SCI shares averages approximately 0.007 across French départements, implying roughly 140 effective UK regions contributing to each département's exposure measure—well above the threshold for many-shock asymptotics.

2. Leave-one-UK-country-out yields stable estimates in the range 0.021–0.026 (**??**), confirming that no single UK constituent country drives the result.

C. Robustness Appendix

C.1 Alternative Clustering

?? reports results under three inference approaches: (i) baseline département-level clustering (SE = 0.011), (ii) two-way clustering by département and quarter-year (SE = 0.011, essentially unchanged), and (iii) heteroskedasticity-robust standard errors (SE = 0.005, much smaller). The coefficient is significant under all three approaches ($p < 0.05$ for clustered, $p < 0.001$ for heteroskedasticity-robust).

C.2 COVID Period Controls

The overlap between the post-transition trade shock and the COVID-19 pandemic is a potential confound. I include an interaction between log SCI to UK and a COVID period indicator (2020–2021) as an additional control. The main Brexit effect survives ($\hat{\beta} = 0.023$, $p = 0.028$), with the COVID interaction absorbing a small, statistically insignificant portion of the variation ($\hat{\beta}^{COVID} = 0.009$, $p > 0.10$).

C.3 Alternative Shock Measures

The baseline uses total SCI (population-weighted proxy) as the exposure measure. **??** shows that probability-weighted SCI (mean SCI) produces nearly identical results ($\hat{\beta} = 0.025$, SE = 0.012), indicating that the cross-sectional variation in total and mean SCI is highly correlated at the département level.

D. Additional Figures and Tables

D.1 Reduced-Form Binscatter

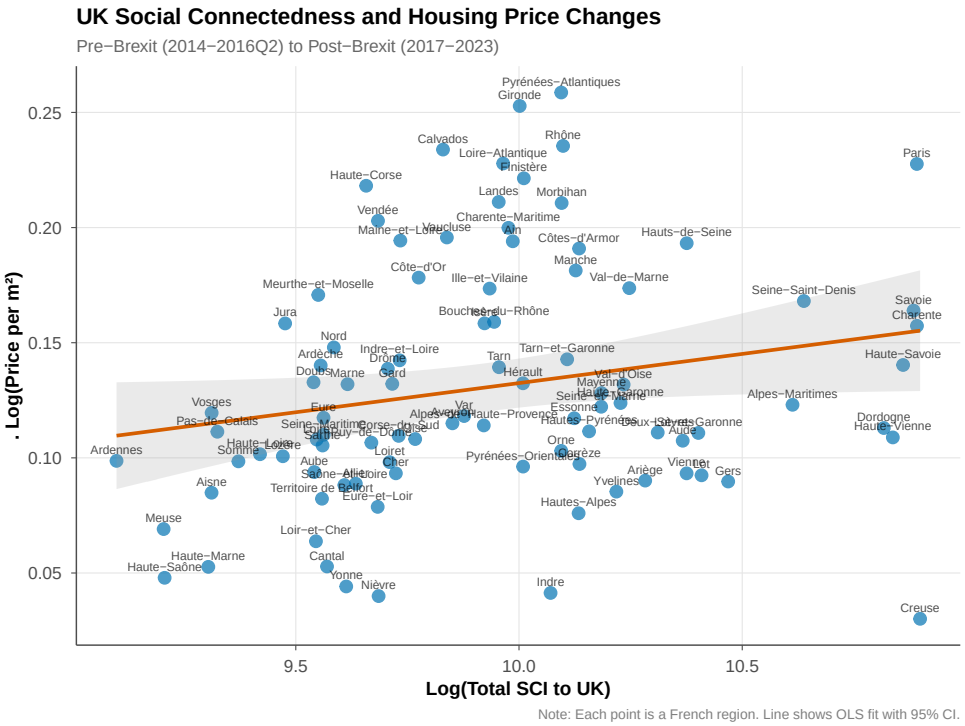

Figure 7: UK Social Connectedness and Housing Price Changes

Notes: Each point is a French département. Horizontal axis: log of total SCI to UK. Vertical axis: change in log price per m^2 from pre-Brexit (2014–2016 Q2) to post-Brexit (2016 Q3–2023 Q4). Line shows OLS fit with 95% CI. The positive slope visualizes the main finding: départements more connected to the UK experienced greater price appreciation.

Table 8: Permutation Inference: Randomization of SCI Exposure

Statistic	Value
Actual coefficient	0.0250
Number of permutations	2000
Mean permuted coefficient	0.0004
SD permuted coefficient	0.0118
Permutation p-value (two-sided)	0.038
Actual / SD(permuted)	2.11

Notes: SCI exposure is randomly reassigned across French regions in each permutation, holding the panel structure and time fixed effects fixed. The p-value is the share of permutations where $|\hat{\beta}_{\text{perm}}| \geq |\hat{\beta}_{\text{actual}}|$. Under the null of no network effect, exposure should not predict outcomes.